

```

rm(list = ls())

# Local github
codepath <- c("/media/larryleon/My Projects/GitHub/Forest-Search/R/")
source(paste0(codepath, "source_forestsearch_v0.R"))
source_fs_functions(file_loc = codepath)

library(kableExtra)
library(knitr)
library(ggplot2)
library(gridExtra)
library(cubature)
library(aVirtualTwins)
library(randomForest)
library(survival)
library(survminer)
library(grf)
library(policytree)
library(data.table)
library(plyr)
library(dplyr)
library(glmnet)
library(corrplot)

maxFollow <- 84
cens.type <- "weibull"

# m1 -censoring adjustment
muC.adj <- log(1.5)
k.z3 <- 1
k.treat <- 0.9

z1_frac <- 0.25 # Default model index 'm1' (The 1st quartile of z1=er)
pH_super <- 0.125 # non-NULL re-defines z1_frac

if (is.null(pH_super)) {
  # pH_check<-with(gbsg,mean(pgr<=quantile(pgr,c(z3_frac),1,0) &
  # er<=quantile(er,z1_frac)))
  pH_check <- with(gbsg, mean(meno == 0 & er <= quantile(er, z1_frac)))
  cat("Underlying pH_super", c(pH_check), "\n")
}
# pH_super specified If pH_super then override z1_frac and find z1_frac to
# yield pH_super

if (!is.null(pH_super)) {
  # Approximate Z1 quantile to yield pH proportion
  z1_q <- uniroot(propH.obj4, c(0, 1), tol = 1e-04, pH.target = pH_super)$root
  # pH_check<-with(gbsg,mean(pgr<=quantile(pgr,c(z3_frac),1,0) &
  # er<=quantile(er,z1_q)))
  pH_check <- with(gbsg, mean(meno == 0 & er <= quantile(er, z1_q)))
  cat("pH", c(pH_check), "\n")
  rel_error <- (pH_super - pH_check)/pH_super
  if (abs(rel_error) >= 0.1)
    stop("pH_super approximation relative error exceeds 10%")
  z1_frac <- z1_q
}

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```

    cat("Underlying pH_super", c(pH_check), "\n")
  }

## pH 0.122449
## Underlying pH_super 0.122449

# Bootstrap on log(hr) scale converted to HR (est.loghr=TRUE & est.scale='hr')
t.start.all <- proc.time()[3]

##### Forest search criteria
hr.threshold <- 1.25 # Initial candidates
hr.consistency <- 1 # Candidates for many splits
pconsistency.threshold <- 0.9
stop.threshold <- 0.95
maxk <- 2
nmin.fs <- 60
pstop_futile <- 0.5
# Limit timing for forestsearch
max.minutes <- 3
m1.threshold <- Inf # Turning this off (Default)
# pconsistency.threshold<-0.70 # Minimum threshold (will choose max among
# subgroups satisfying)
fs.splits <- 400 # How many times to split for consistency
# vi is % factor is selected in cross-validation --> higher more important
vi.grf.min <- 0.2
# Null, turns off grf screening
d.min <- 10 # Min number of events for both arms (d0.min=d1.min=d.min)
# default=5 Virtual twins analysis Counter-factual difference (C-E) >=
# vt.threshold Large values in favor of C (control)
vt.threshold <- 0.225 # For VT delta
treat.threshold <- 0

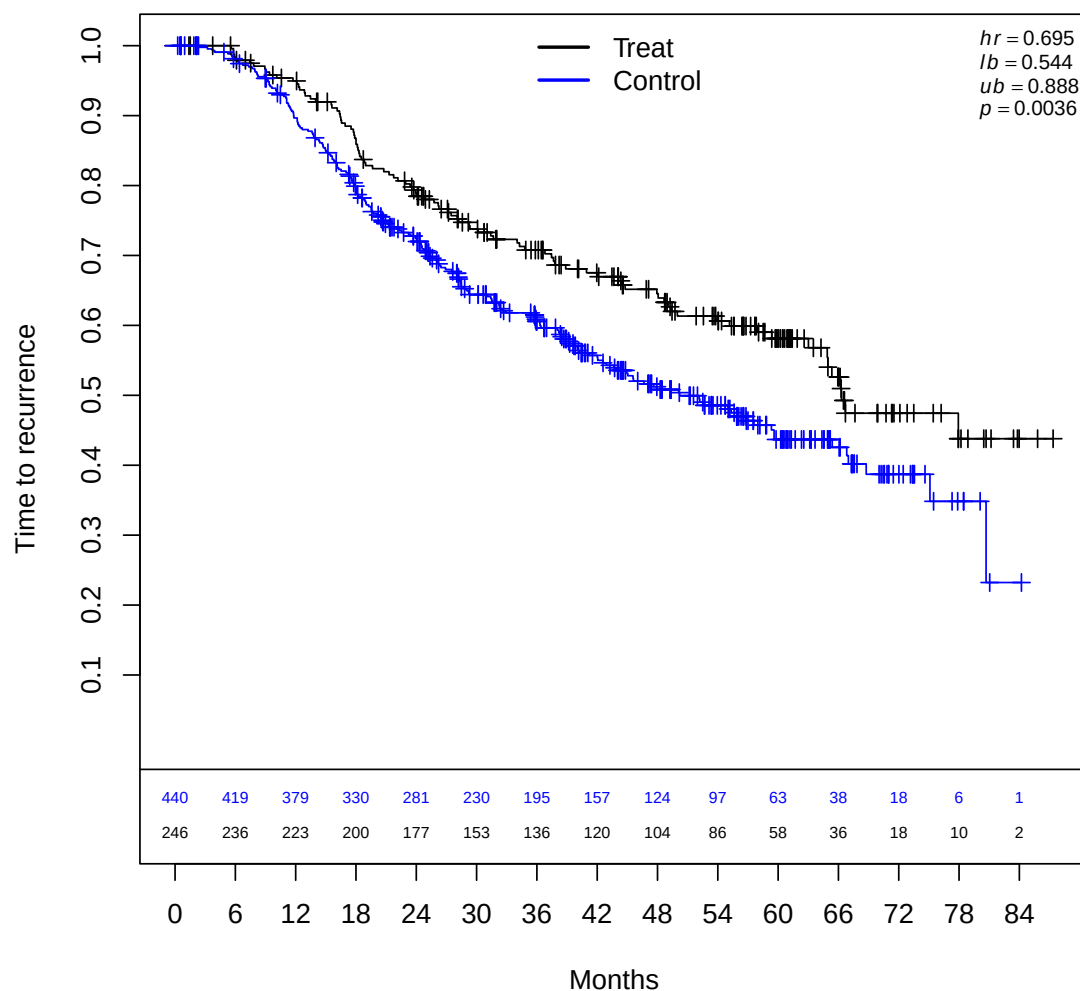
maxdepth <- 2
n.min <- 60
ntree <- 1000
# GRF criteria
dmin.grf <- 12 # For GRF delta
# Note: For CRT this represents dmin.grf/2 RMS for control (-dmin.grf/2 for
# treatment)
frac.tau <- 0.6

outcome.name <- c("y.sim")
event.name <- c("event.sim")
id.name <- c("id")
treat.name <- c("treat")

cox.formula.sim <- as.formula(paste("Surv(y.sim,event.sim)~treat"))
cox.formula.adj.sim <- as.formula(paste("Surv(y.sim,event.sim)~treat+v1+v2+v3+v4+v5"))

mod.harm <- "alt"
hrH.target <- 2.5
# out.loc = NULL turns off file creation
N <- 700
this.dgm <- get.dgm4(mod.harm = mod.harm, N = N, k.treat = k.treat, hrH.target = hrH.target,
  cens.type = cens.type, out.loc = NULL, details = TRUE, parms_torand = FALSE)

```



```
## Super-population empirical harm and non-harm hazard ratios= 2.499996 0.6466405
## Causal HR (empirical ITT)= 0.7125876
```

```
file_out <- c("output/sim-FS4_Lasso_ex5.Rdata")

n_add_noise <- 3

# sim=69 sim=71

sim <- 39

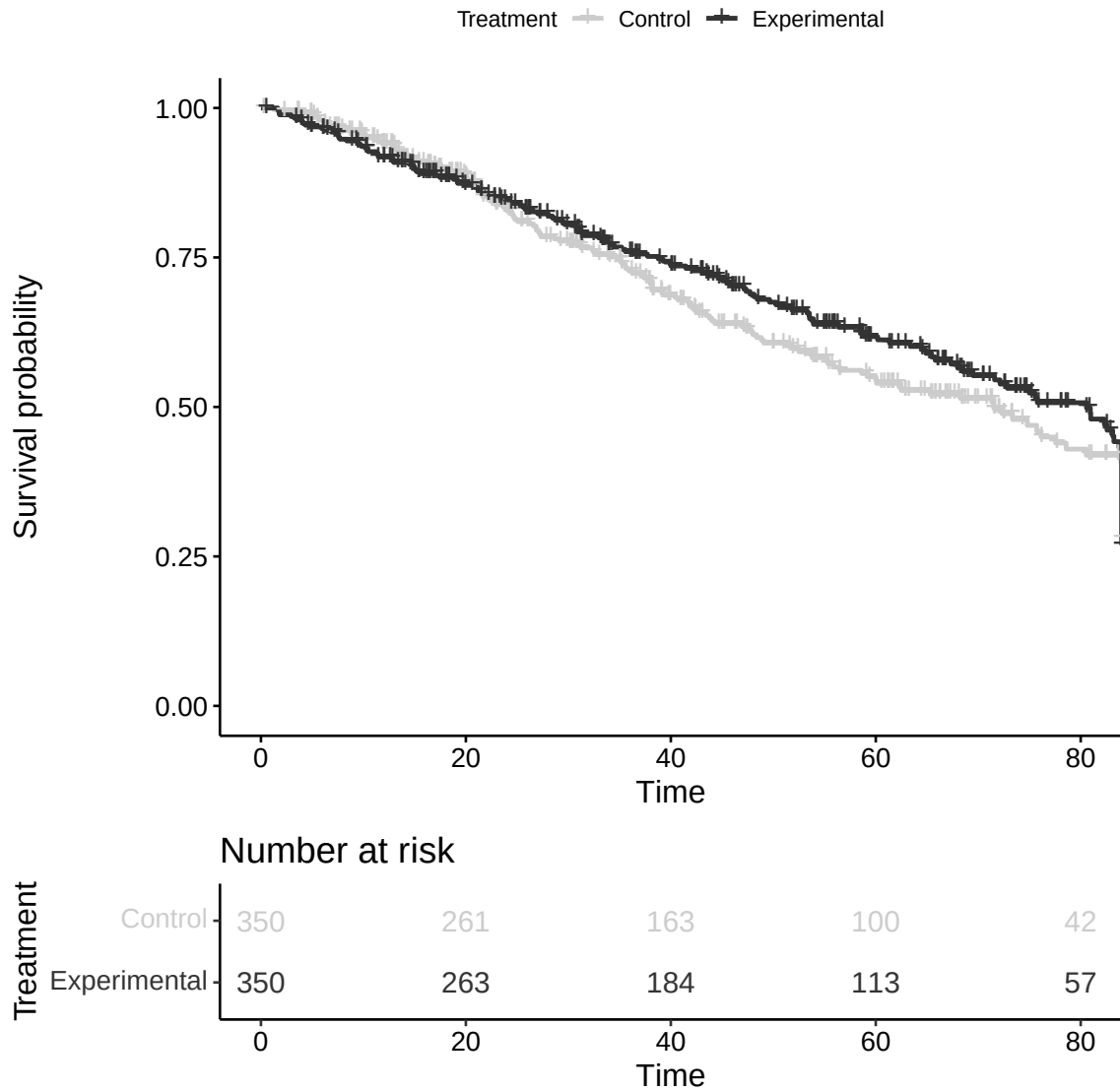
dgm <- this.dgm$dgm

x <- sim_aftm4_gbsg(dgm = dgm, n = N, maxFollow = maxFollow, muC.adj = muC.adj, simid = sim)

kmfit <- survfit(Surv(y.sim, event) ~ treat, data = x)

ggsurvplot(kmfit, data = x, main = "K-M curves for simulated data", legend = "top",
```

```
legend.title = "Treatment", legend.labs = c("Control", "Experimental"), palette = "grey",
risk.table = TRUE, risk.table.col = "strata")
```



```
if (n_add_noise == 0) {
  confounders.name <- c("z1", "z2", "z3", "z4", "z5", "size", "grade")
}

if (n_add_noise == 5) {
  set.seed(8316951 + 1000 * sim)
  # Add 5 noise
  x$noise1 <- rnorm(N)
  x$noise2 <- rnorm(N)
  x$noise3 <- rnorm(N)
  x$noise4 <- rnorm(N)
  x$noise5 <- rnorm(N)
  confounders.name <- c("z1", "z2", "z3", "z4", "z5", "size", "grade", "noise1",
    "noise2", "noise3", "noise4", "noise5")
}
```

```

}

if (n_add_noise == 3) {
  set.seed(8316951 + 1000 * sim)
  # Add 3 noise
  x$noise1 <- rnorm(N, sd = 1)
  x$noise2 <- rnorm(N, sd = 1)
  x$noise3 <- rnorm(N, sd = 1)
  confounders.name <- c("z1", "z2", "z3", "z4", "z5", "size", "grade", "noise1",
    "noise2", "noise3")
}

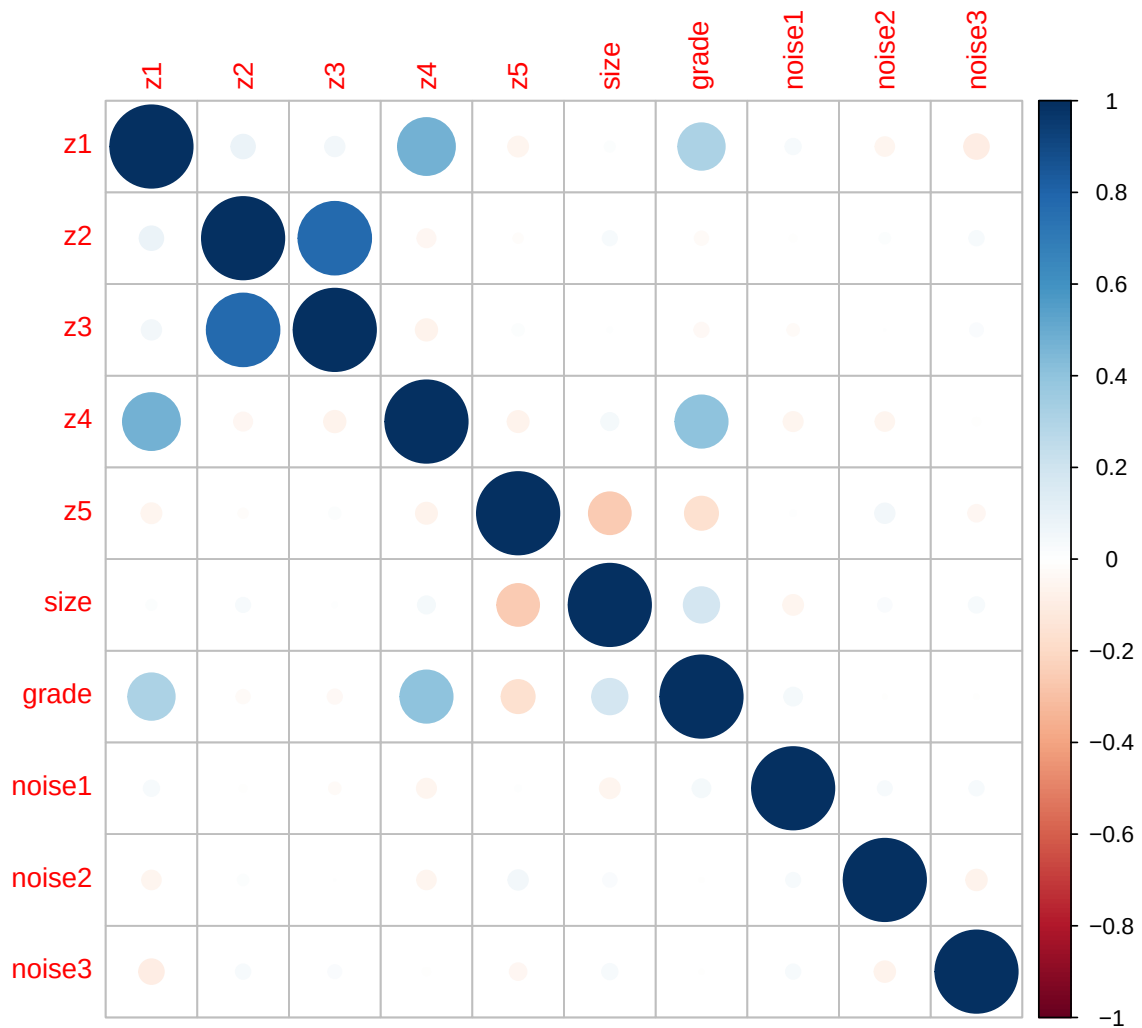
# More challenging which_replace <- which(confounders.name=='z1')
# confounders.name[which_replace] <- 'er'

cox.formula.check <- as.formula(paste("Surv(y.sim,event.sim)~treat+v1+v2+v3+v4+v5+size+grade+noise1+
coxph(cox.formula.check, data = x)

## Call:
## coxph(formula = cox.formula.check, data = x)
##
##              coef exp(coef) se(coef)      z      p
## treat  -0.246451  0.781570  0.105979 -2.325  0.02005
## v11     0.354916  1.426061  0.128803  2.755  0.00586
## v21    -0.083822  0.919595  0.169012 -0.496  0.61993
## v31     0.043540  1.044502  0.172191  0.253  0.80038
## v41     0.593541  1.810388  0.127329  4.661 3.14e-06
## v51    -0.902570  0.405526  0.109274 -8.260 < 2e-16
## size    0.004553  1.004563  0.003506  1.298  0.19413
## grade   0.112937  1.119561  0.096212  1.174  0.24046
## noise1  0.013467  1.013558  0.055415  0.243  0.80799
## noise2 -0.056197  0.945353  0.053447 -1.051  0.29305
## noise3 -0.102944  0.902177  0.057695 -1.784  0.07438
##
## Likelihood ratio test=162.1 on 11 df, p=< 2.2e-16
## n= 700, number of events= 382

Zm <- cor(as.matrix(x[, c(confounders.name)]))
corrplot(Zm)

```



```
# Options Allconfounders.name is list of confounders within analysis dataset
# (1) use_lasso=TRUE & use_grf=FALSE Lasso used to possibly reduce dimension
# Any continuous factors are cut at medians (2) use_lasso=TRUE & use_grf=TRUE
# Lasso used to reduce dimension Continuous covariates are cut at medians
# However, if GRF selects a covariate cut then only that cut is used: For
# example if 'age <= median(ag)' is called for per Lasso but GRF includes 'age
# <= 54', then only the latter is used (3) use_grf_only = TRUE (overrides
# use_lasso and use_grf) Only factors selected via GRF are used (4) use_lasso =
# F & use_grf =T Median cuts (unless selected via GRF) as in (2) However no
# possible dimension reduction via lasso All categorical factors included

use_lasso <- TRUE
use_grf <- FALSE
use_grf_only <- FALSE

fs.est <- forestsearch(df.analysis = x, Allconfounders.name = confounders.name, details = TRUE,
  use_lasso = use_lasso, use_grf = use_grf, use_grf_only = use_grf_only, dmin.grf = 12,
  frac.tau = 0.6, conf_force = NULL, outcome.name = outcome.name, treat.name = treat.name,
```

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event.name = event.name, id.name = id.name, n.min = nmin.fs, hr.threshold = hr.threshold,
hr.consistency = hr.consistency, fs.splits = fs.splits, d0.min = d.min, d1.min = d.min,
pstop_futile = pstop_futile, pconsistency.threshold = pconsistency.threshold,
stop.threshold = stop.threshold, max.minutes = max.minutes, maxk = maxk, by.risk = 12,
plot.sg = TRUE, vi.grf.min = vi.grf.min)

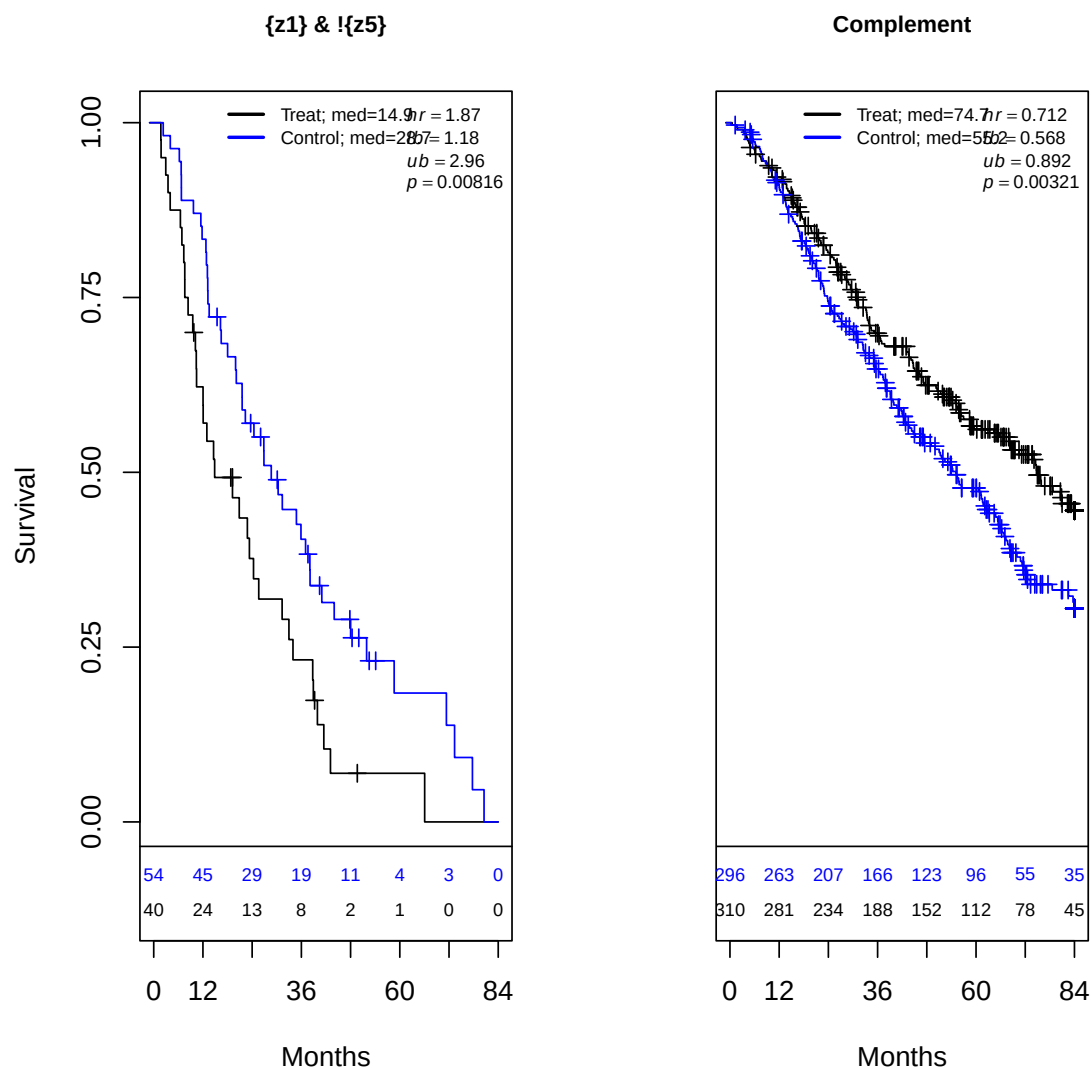
## # of continuous/categorical characteristics 4 6
## Continuous characteristics: size noise1 noise2 noise3
## Categorical characteristics: z1 z2 z3 z4 z5 grade
## 10 x 2 sparse Matrix of class "dgCMatrix"
##           s0      s0
## z1      0.365077995 0.427304741
## z2      .          .
## z3      .          .
## z4      0.499835997 0.585031950
## z5     -0.854373846 1.000000000
## size    0.001662036 0.001945326
## grade   0.094953403 0.111138003
## noise1  .          .
## noise2 -0.010752306 0.012585013
## noise3 -0.047485212 0.055578962
## Cox-LASSO selected: z1 z4 z5 grade noise3
## Cox-LASSO not selected: z2 z3 size noise1 noise2
## Median cuts (after Lasso if use_lasso) noise3
## Categorical (after Lasso if use_lasso) z1 z4 z5 grade
## Prior to removing (if in GRF): noise3
## Left (to cut at median) after removing others if in GRF: noise3
## # of candidate subgroup factors= 5
## [1] "noise3 <= median(noise3)" "z1"
## [3] "z4"                        "z5"
## [5] "grade"
## Confounders per grf screening q2 q5 q1 q4 q3
## FSconfounders.name      vi.cs vi_ratio
## 2          q2 0.33034292 1.6517146
## 5          q5 0.27188139 1.3594069
## 1          q1 0.17193143 0.8596571
## 4          q4 0.13547259 0.6773629
## 3          q3 0.09037167 0.4518584
## Number of unique levels (L) and possible subgroups= 11 2047
## # of subgroups based on # variables > k.max and excluded (per million) 0.001981
## k.max= 2
## Events criteria for control,exp= 10 10
## # of subgroups with events less than criteria: control, experimental 12 12
## # of subgroups with sample size less than criteria 14
## # of subgroups meeting all criteria = 52
## # of subgroups fitted (Cox model estimable) = 52
## *Subgroup Searching Minutes=* 0.002383333
## Number of subgroups meeting HR threshold 5
## # of candidate subgroups (meeting HR criteria) = 5
## Subgroups (1st 10) meeting overall screening thresholds (HR, m1) sorted by HRs= Inf
##      K  n  E d1  m1  m0  HR L(HR) U(HR) q2.0 q2.1 q5.1 q5.2 q5.3 q1.0
## 1:  2  94 78 35 14.85 28.66 1.87 1.18 2.96  0  1  0  0  0  0
## 2:  2  91 61 30 30.74 47.38 1.51 0.91 2.51  0  0  0  0  1  1
## 3:  2  89 58 29 31.32 44.30 1.42 0.84 2.38  1  0  0  0  1  0
## 4:  2 100 84 42 19.20 26.85 1.38 0.89 2.14  0  0  0  0  1  0
## 5:  2  79 55 27 31.31 44.01 1.32 0.78 2.25  0  1  0  0  0  1

```

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## 6: NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## 7: NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## 8: NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## 9: NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
## 10: NA NA NA NA NA NA NA NA NA NA NA NA NA NA NA
##      q1.1 q4.0 q4.1 q3.0 q3.1
## 1:      0      1      0      0      0
## 2:      0      0      0      0      0
## 3:      0      0      0      0      0
## 4:      0      1      0      0      0
## 5:      0      0      0      0      0
## 6:      NA      NA      NA      NA      NA
## 7:      NA      NA      NA      NA      NA
## 8:      NA      NA      NA      NA      NA
## 9:      NA      NA      NA      NA      NA
## 10:     NA      NA      NA      NA      NA
## Consistency 0.9975
## # of splits= 400
## Model, % Consistency Met= {z1} !{z5} 0.9975
## Number of subgroups meeting consistency criteria=
##      p.consistency Nsg group.id m.index K   M.1   M.2
## 1:      0.9975  94      5      1 2 {z1} !{z5}

```



```
## [1] "{z1}" "!("{z5}"
## % consistency criteria met= 0.9975
## SG focus= hr
## Subgroup Consistency Minutes= 0.02315
## Subgroup found
## Minutes overall= 0.03728333

grf.est <- grf.subg.harm.survival(data = x, confounders.name = confounders.name,
  outcome.name = outcome.name, event.name = event.name, id.name = id.name, treat.name = treat.name,
  n.min = n.min, dmin.grf = dmin.grf, frac.tau = 1, details = TRUE)

## tau, maxdepth= 83.73135 2
## leaf.node control.mean control.size control.se treated.mean treated.size
## 1 2 -11.345975 396.000000 2.703771 11.345975 396.000000
## 2 3 4.460267 304.000000 3.217394 -4.460267 304.000000
## 3 4 -14.517993 176.000000 4.052323 14.517993 176.000000
## 4 5 8.499669 79.000000 5.776644 -8.499669 79.000000
## 5 6 -11.954760 247.000000 3.435930 11.954760 247.000000
## 6 7 8.582998 198.000000 4.027162 -8.582998 198.000000
```

```
##   treated.se      diff Nsg depth
## 1   2.703771 -22.691951 396     1
## 2   3.217394   8.920533 304     1
## 3   4.052323 -29.035987 176     2
## 4   5.776644  16.999337  79     2
## 5   3.435930 -23.909521 247     2
## 6   4.027162  17.165996 198     2
##   leaf.node control.mean control.size control.se treated.mean treated.size
## 6         7      8.582998   198.000000   4.027162   -8.582998   198.000000
##   treated.se      diff Nsg depth
## 6   4.027162 17.166 198     2
## [1] "size <= 23" "z1 <= 0"   "z3 <= 0"
## [1] "z3 <= 0"
```

```
library(doParallel)
registerDoParallel(parallel::detectCores(logical = FALSE))

cox.formula.boot <- cox.formula.sim
max.minutes <- 7
# Suggest running 50, first ... to get timing estimate

NB <- 1000

df_boot_analysis <- fs.est$df.est

fitH <- get_Cox_sg(df_sg = subset(df_boot_analysis, treat.recommend == 0), cox.formula = cox.formula)
H_obs <- fitH$est_obs # log(hr) scale
seH_obs <- fitH$se_obs
# Hc observed estimates
fitHc <- get_Cox_sg(df_sg = subset(df_boot_analysis, treat.recommend == 1), cox.formula = cox.formula)
Hc_obs <- fitHc$est_obs
seHc_obs <- fitHc$se_obs
rm("fitH", "fitHc")

Ystar_mat <- bootYstar({
  ystar <- get_Ystar(boot)
}, boots = NB, seed = 8316951, counter = "boot", export = fun_arg_list_boot)
# Check dimension
if (dim(Ystar_mat)[1] != NB | dim(Ystar_mat)[2] != nrow(df_boot_analysis)) stop("Dimension of Ystar_mat")

tB.start <- proc.time()[3]
# Bootstraps
resB <- bootPar({
  ans <- fsboot_forparallel(boot)
}, boots = NB, seed = 8316951, counter = "boot", export = fun_arg_list_boot)
tB.now <- proc.time()[3]
tB.min <- (tB.now - tB.start)/60

doParallel::stopImplicitCluster()

cat("Minutes for Boots", c(NB, tB.min), "\n")

## Minutes for Boots 1000 1.371583

cat("Projection per 1000", c(tB.min * (1000/NB)), "\n")
```

```

## Projection per 1000 1.371583

cat("Propn bootstrap subgroups found =", c(sum(!is.na(resB$H_biasadj_1))/NB), "\n")

## Propn bootstrap subgroups found = 0.935

# How many timmed out
cat("Number timmed out=", c(sum(is.na(resB$H_biasadj_1) & resB$tmmins_search > max.minutes)),
    "\n")

## Number timmed out= 0

H_estimates <- get_dfRes(Hobs = H_obs, seHobs = seH_obs, H1_adj = resB$H_biasadj_2,
    ystar = Ystar_mat, cov_method = "standard", cov_trim = 0)
Hc_estimates <- get_dfRes(Hobs = Hc_obs, seHobs = seHc_obs, H1_adj = resB$Hc_biasadj_2,
    ystar = Ystar_mat, cov_method = "standard", cov_trim = 0)

print(H_estimates)

##           H0          sdH0 H0_lower H0_upper          H1          sdH1 H1_lower H1_upper
## 1: 1.866678 0.4404354 1.175525 2.964198 1.591189 0.3916811 0.9821851 2.577807

print(Hc_estimates)

##           H0          sdH0 H0_lower H0_upper          H1          sdH1 H1_lower
## 1: 0.7117625 0.08211457 0.5677194 0.8923526 0.7191694 0.1217999 0.5160241
##           H1_upper
## 1: 1.002288

# H_estimates<-get_dfRes(Hobs=H_obs,seHobs=seH_obs,H1_adj=resB$H_biasadj_1,ystar=Ystar_mat,cov_metho
# Hc_estimates<-get_dfRes(Hobs=Hc_obs,seHobs=seHc_obs,H1_adj=resB$Hc_biasadj_1,ystar=Ystar_mat,cov_m
# print(H_estimates) print(Hc_estimates)

bootit <- list(H_estimates = H_estimates, Hc_estimates = Hc_estimates)
ans3 <- fsBoot.parallel.out(df = df_boot_analysis, FSboots = bootit, cox.formula.boot = cox.formula.
    sim_number = sim)
tall.min <- (tB.now - t.start.all)/60
cat("Overall minutes for analysis", c(tall.min), "\n")

## Overall minutes for analysis 1.497367

save(dgm, x, fs.est, grf.est, bootit, ans3, tall.min, resB, cox.formula.boot, file = file_out)

## Confounders evaluated in Forestsearch:
## [1] "noise3 <= median(noise3)" "z1"
## [3] "z4" "z5"
## [5] "grade"

```

Table 1 Simulated data example: Cox hazard ratio (hr) estimates for the ITT population and subgroups H and H^c . Cox model estimates are based on subgroups: H true (knowing the actual subgroup, a-priori); the estimated subgroup $\hat{H}$; and the bootstrap (1,000 resamples) bias-correction to $\hat{H}$ estimates, denoted $\hat{H}_{bc}$. Estimates for the complement H^c are defined analogously. The number of subjects in each population (# Subjects) and the number of subjects correctly classified (# Correct) in the true subgroups H and H^c , respectively, are listed.

	HR Estimate	Lower	Upper	# Subjects	# Correct
ITT					
ITT	0.762	0.623	0.932	700	.
H subgroup estimates					
H_{oracle}	2.296	1.368	3.855	89	.
$\hat{H}$	1.867	1.176	2.963	94	50
$\hat{H}_{bc}$	1.591	0.982	2.579	94	50
H-complement subgroup estimates					
H_{oracle}^c	0.664	0.532	0.829	611	.
$\hat{H}^c$	0.712	0.568	0.892	606	567
$\hat{H}_{bc}^c$	0.719	0.516	1.003	606	567